ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** push or a pull	Forces are pushes or pulls.
2	**(d)** rough surfaces	Rough surfaces interlock.
3	**(b)** positive and negative	Two kinds of charge.
4	**(c)** newton	Named after Isaac Newton.
5	**(d)** lubricants	Oil forms a layer.
6	**(d)** repel each other	Same charges repel.
7	**(b)** the speed, direction or shape of an object	Effects of force.
8	**(d)** less	Rolling friction is smaller than sliding friction.
9	**(b)** attract each other	Opposite attract.
10	**(c)** friction	Friction acts only on contact.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A push or a pull (1); Newton (N) (1).	1+1
12	To reduce friction (1); So the coins slide easily (1).	1+1
13	Like charges repel (1); Unlike charges attract (1).	1+1
14	Contact: muscular force or friction (1); Non-contact: gravity or magnetic force (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Force can squeeze, stretch or bend (1); Pressing dough, stretching a rubber band (1); Shape change depends on force and material (1).	1+1+1
16	A metal rod with two thin leaves in a jar (1); Touching a charged body spreads charge to leaves (1); Like charges repel; leaves diverge (1).	1+1+1
17	Net force = 0 (1); Forces are balanced (1); The book stays at rest (1).	1+1+1
	OR: Static: pushing a heavy box that does not move (1); Sliding: a box sliding on the floor (1); Rolling: a ball rolling on the ground (1).	1+1+1
18	Rubbing transfers charges from one body to another (1); One gains negative charge; other becomes positive (1); A glass rod rubbed with silk (1).	1+1+1
	OR: Bring a magnet near iron, wood, plastic, steel (1); Iron and steel are attracted (1); Wood and plastic are not (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Moves a stationary body: kicking a ball (1); Stops a moving body: catching a ball (1); Changes direction: hockey stick hits (1); Changes speed: pedalling faster (1); Changes shape: squeezing clay (1).	1+1+1+1+1
	OR: Leaves hang together when uncharged (1); Touching with a charged rod gives charge to all parts (1); Leaves carry the same charge and repel (1); More charge, more divergence (1); Used to detect charge (1).	1+1+1+1+1
20	Contact: muscular, friction (1); Non-contact: gravitational, magnetic (1); Electrostatic force acts at a distance (1); Contact forces need touching (1); Non-contact forces act through space (1).	1+1+1+1+1
	OR: Earth attracts all objects (1); Weight is gravitational force (1); Moon's gravity is smaller (1); Weight decreases on the Moon (1); Mass does not change (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.

- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** change sliding into rolling friction	Reduce friction.
2	**(c)** gold-leaf electroscope	Leaves diverge.
3	**(a)** gravitational force	Acts at a distance.
4	**(b)** heat	Rubbing hands warm them.
5	**(b)** a charged body touches the metal disc	Same charges repel.
6	**(b)** electrostatic force	Charged objects attract light things.
7	**(c)** an object is at rest but tends to move	Prevents the start of motion.
8	**(a)** discharge of electric charges between clouds or with the earth	Static charge discharge.
9	**(b)** gravitational force	Gravity.
10	**(c)** sliding friction	Sliding.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To increase friction with the road (1); Prevents skidding (1).	1+1
12	Earth's gravitational force pulls it down (1); Gravity acts on all objects (1).	1+1
13	It can change speed or direction of motion (1); It can change the shape of the object (1).	1+1
14	Equal forces in opposite directions on a body (1); Net force is zero (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lubricants like oil and grease (1); Ball bearings (1); Making surfaces smooth (1).	1+1+1
16	Fires, damage to buildings, and loss of life (1); Stay indoors, avoid tall trees (1); Use lightning conductors; stay away from water (1).	1+1+1
17	Water or oil reduces friction (1); Foot slips on the floor (1); Reduced grip leads to falls (1).	1+1+1
	OR: Roughening surfaces (1); Treads on tyres (1); Using sand or ash on slippery ground (1).	1+1+1
18	Force of attraction between objects having mass (1); Fall of an apple (1); Moon going round the Earth (1).	1+1+1
	OR: Friction opposes motion; gravity pulls to Earth (1); Friction needs contact (1); Gravity acts at a distance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pull a block on smooth and rough surfaces with a spring balance (1); Reading is larger on rough surfaces (1); Add weights on the block (1); Reading rises with load (1); Friction depends on both (1).	1+1+1+1+1
	OR: Balanced forces: no change in motion (1); Unbalanced: speed or direction changes (1); Shape changes by squeezing, bending (1); Example: tug of war (1); Net force decides (1).	1+1+1+1+1
20	Helpful: walking, brakes, grip (1); Harmful: wear, heat, loss of energy (1); Reduce: oil, bearings (1); Increase: treads, rough surfaces (1); Balance as needed (1).	1+1+1+1+1
	OR: Static > sliding > rolling (1); Wheels and bearings use rolling (1); Suitcases have wheels (1); Less force needed (1); Less wear (1).	1+1+1+1+1

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- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.

- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** passing the charge safely to the earth	A metal rod connected to ground.
2	**(d)** opposite to the direction of motion or tendency of motion	It opposes motion.
3	**(b)** wearing out of shoe soles	Wears surfaces.
4	**(c)** positively charged	Glass loses electrons.
5	**(a)** balanced	Net force is zero.
6	**(c)** we can walk without slipping	Grip.
7	**(d)** iron	Magnetic material.
8	**(d)** spring balance	Spring extension shows force.
9	**(b)** static friction equal to the push	Balanced forces.
10	**(a)** repel	Like poles repel.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Protect buildings from lightning (1); Carries charge to the ground (1).	1+1
12	Force opposing relative motion between surfaces (1); Due to irregularities of the two surfaces in contact (1).	1+1
13	It attracts the bits (1); Rubbing charges the scale; electrostatic force acts (1).	1+1
14	Spring stretches more for a larger force (1); Scale reads the force in newton (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	We can walk and hold objects (1); Brakes stop vehicles (1); Matchstick lights (1).	1+1+1
16	Mass is the quantity of matter; weight is gravitational force (1); Mass in kg; weight in newton (1); Mass is the same everywhere; weight changes with place (1).	1+1+1
17	Force due to the action of muscles (1); Lifting a bucket (1); Pulling a cart, kicking a ball (1).	1+1+1
	OR: Wear and tear of machines (1); Wastes energy as heat (1); Slows moving objects (1).	1+1+1
18	Static friction is greater than sliding friction (1); More force is needed to overcome static friction (1); Once moving, sliding friction is smaller (1).	1+1+1
	OR: Air pushes against the motion (1); Shapes of cars and aeroplanes reduce drag (1); Parachute increases drag (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rubbing the balloon charges it (1); It attracts bits of paper (1); Two charged balloons of the same charge repel (1); Opposite charges attract (1); Charges are positive and negative (1).	1+1+1+1+1
	OR: Like poles repel (1); Unlike poles attract (1); Force acts without touching (1); Float magnet above another by like poles (1); Used in maglev trains (1).	1+1+1+1+1
20	Charge builds up in clouds (1); Discharge causes lightning (1); Conductor sends charge to earth (1); Stay indoors, avoid trees (1); Avoid water and metal objects (1).	1+1+1+1+1
	OR: Pressing: dough (1); Pulling: drawer (1); Stretching: rubber band (1); Twisting: wringing cloth (1); Different forces give different effects (1).	1+1+1+1+1

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MASTERMIND